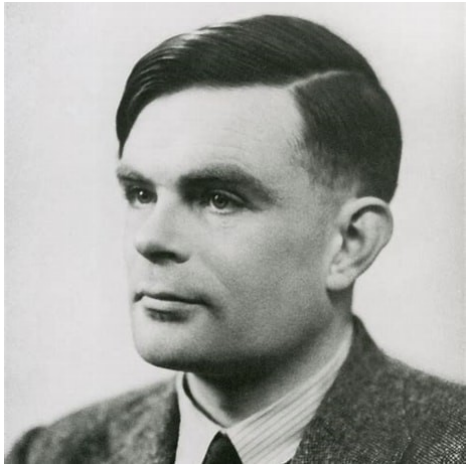
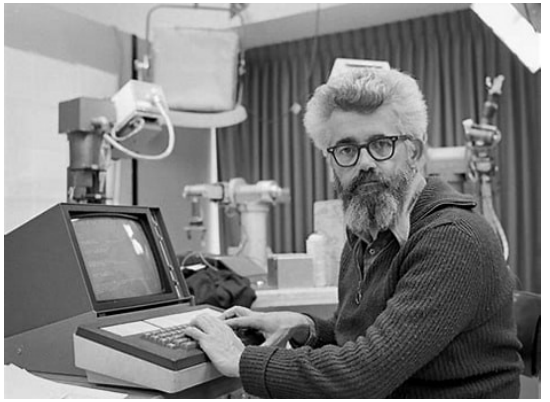
AI Events and People 1950s

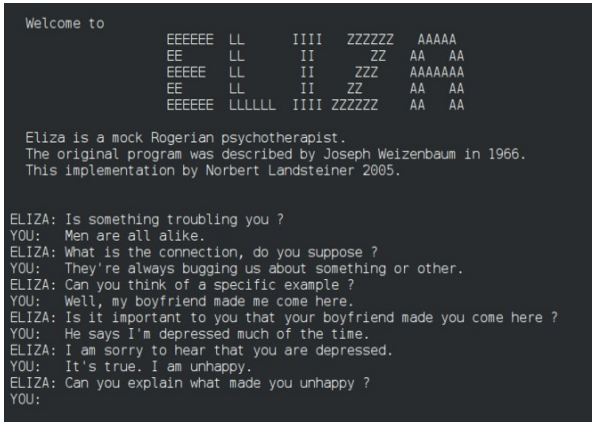
- (1912-1954) Alan Turing
  - Theoretical father of CS. “Turing Test”
  - Early thoughts on “connectionist” AI – forerunner to deep learning
- (1956) “Dartmouth conference”
  - John McCarthy coins “Artificial Intelligence”
- (1956) Logic Theorist
  - First good “theorem prover” – symbolic AI





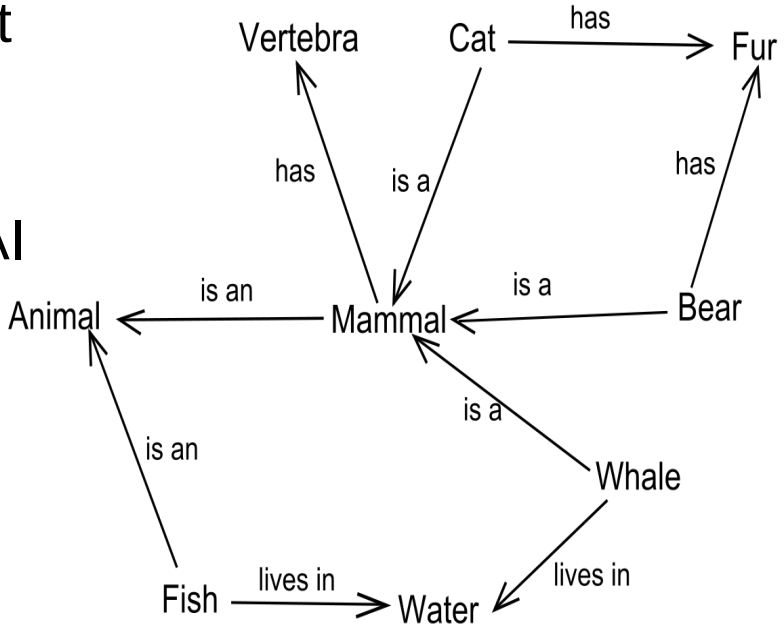
AI Events and People 1960s

- (1965) ELIZA
  - Very basic “chatbot” in 200 lines of code
- (1967) Minsky: “Within a generation … the problem of creating artificial intelligence will substantially be solved.”
- (1968) First chess-playing program able to beat a decent human player
- (1969) Minsky and Papert publish “Perceptrons”
  - Proves early neural networks very limited
  - One of the triggers of 1970s “AI winter”



AI People and Events 1970-1985

- First “AI Winter”
- (1974) Paul Werbos PhD describes neural net backpropagation algorithm in detail. No one notices.
- Gradual development of symbol/logic based AI
- (1982) Japan announces “5th generation computing” project – hardware devoted to expert systems or symbolic reasoning
- Various *limited* success with expert systems



AI People and Events 1985-1990

- (1986) Backpropagation algorithm rediscovered. Neural nets are back, briefly.
  - But training > 3 layers proves very hard.
- Continued work on symbolic AI and expert systems.
  - But, symbolic AI is limited..
  - .. and fragile
  - .. and 5th generation project peters out

AI People and Events 1990-2012

- Second “AI winter”
- Gradual improvements in neural networks, by Geoff Hinton, Ilya Sutskever, Yan LeCunn, Yoshua Bengio and many others.
- Other pattern recognition methods, e.g. SVMs, explored.
- Various limited neural net successes e.g. character recognition for written digits.

AI Events 2012-2017

- Four important advances lay ground for new neural network growth
  - more available training data via internet
  - computing power via GPUs
  - Better neural twork design: “activation functions” and architectures (neural net => deep learning)
  - Vibrant research community
- (2012) Surprise success of neural networks in image recognition
- (2015) “LSTM” architecture improves language processing
- (2017) Alpha Go beats world Go champion

Last 7 years

- (2017) Vaswani ***et al*** publish “All you Need is Attention”
- (2020, 2022, 2023) OpenAI releases GPT-3, GPT-3.5, GPT-4
- (2020) Deep learning solves protein folding
- (2020 – 2024) Diffusion models generate images and movies

AI Events 2025

- Nobel Prizes
  - Physics to John Hopfield and Geoffrey Hinton
  - Chemistry to the developers of Alpha Fold
- Improved LLM Architectures
  - MoE – “Mixture of Experts”
  - CoT – “Chain of Thought”
- DeepSeek
  - Chinese innovators
  - Demonstrates much more efficient use of computation

AI History – Big Ideas

- Early investigation of major ideas
  - Neural networks since 1960s
  - Early “chatbots” in 1960s
- Alternating optimism and AI winters
  - Easy to underestimate the challenge
  - Repeated pattern of giving up on neural networks prematurely
- Symbolic logic vs Pattern Recognition
  - The former looked promising; the latter worked.
  - “Bitter Lesson” – don’t program AI directly; let it self-organize

AI Future – Where is it heading?

- "… heavier than air flying machines are impossible"
  - Lord Kelvin, 1895
- Alchemistic Empiricism
  - Analogy of chemistry in 1825
  - Results in fits and starts as we try things out by guess-and-by-golly
- Start with Nature, End with Design
  - Planes vs birds
  - Deep Learning vs brains : Geoff Hinton's warning

What Does History Tell Us Now?

- LLMs are only part of the story
- Expect some period of slowdown / disillusion
  - Very often true for exciting new tech, e.g. internet in 1999
- But, long term, expect continued progress
  - Each “winter”, short or long, was followed by big jump forward
- Big advances may result from:
  - Better theoretical understanding
  - Improvements on nature

Future Predictions – Dystopian SF

- Premise: Technology is a power unto itself
- Blade Runner (1984)
  - LA is stygian hellscape in 2020s
- Terminator (1984)
  - LA is nuclear-blasted wreck with robots killing humans in 2020s
- Colossus (1970)
  - AI computer becomes world dictator in late 1970s

Future Predictions – CS Perspective

- Premise: Technology is Mind's creativity unfolding.
- "The mariner will have dominion over the atmosphere and the great deep .. the astronomer will no longer look up to the stars, – he will look out from them.." p125: 25-29

